Training Parts vs. Matching Parts: A Controlled Study of Pose-Based Gait Recognition

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Abstract

Local anatomical features can contribute to gait retrieval without receiving a separate identity objective. We examine this distinction on two-dimensional pose sequences from the Chinese Academy of Sciences' Institute of Automation gait dataset B (CASIA-B). A controlled study trains one multi-stream graph backbone under six objectives and three seeds on identity-disjoint development data. Adding part supervised contrastive learning reduces mean Rank-1 under synthetic joint missingness from 63.90% to 57.78%; the paired difference is $\mathbf{-6.12}$ percentage points with a crossed 95% bootstrap interval of $[\mathbf{-10.29}, \mathbf{-2.00}]$. In frozen baseline checkpoints, replacing untrained local projections with normalized raw part pools raises this endpoint to 68.23%, a gain of $\mathbf{+4.32}$ points, $[\mathbf{+2.89}, \mathbf{+5.71}]$. Ten random-head controls and a uniform-weight ablation distinguish this matching benefit from projection initialization and confidence gating. With a fixed recipe, five full-training runs increase standard-protocol Rank-1 from $\mathbf{48.35 \pm 1.85}$% to $\mathbf{54.04 \pm 2.18}$% on the official 50-identity test split. This split was examined in earlier project experiments, so the result demonstrates within-dataset transfer rather than independent confirmation. Uniform and reliability-weighted raw matching attain the same average. The results support evaluating local matching separately from local supervision; their extension to stronger backbones, other datasets, and real occlusion remains untested.

Keywords: gait recognition, 2D pose, occlusion robustness, supervised contrastive learning, part-based matching, negative results

1 Introduction

Pose-based gait recognition retrieves identities from joint trajectories, offering a compact representation with reduced dependence on clothing, background, and illumination (Liao, Yu, An, & Huang, 2020). Graph encoders such as ST-GCN, GaitGraph, and GaitGraph2 make the spatial and temporal structure of this input explicit (Teepe, Gilg, Herzog, Hörmann, & Rigoll, 2022; Teepe et al., 2021; Yan, Xiong, & Lin, 2018). However, recognition depends on the evidence recovered by the pose estimator. Occlusion and detector failures can remove joints or produce incorrect tracks; partial body evidence is consequently a central challenge (P. Huang et al., 2024; T. Li et al., 2024).

Anatomical features offer two distinct interventions. During training, a local identity objective can encourage each body region to discriminate individuals. During retrieval, local similarities can supplement a global sequence score. These interventions are often bundled, although a region that provides useful complementary matching evidence need not be independently discriminative under a local loss. Supervised contrastive learning supplies a plausible local objective (Khosla et al., 2020), but its benefit for gait parts is an empirical question. Separating training from matching is therefore necessary for attributing any improvement to the intended intervention.

We examine this distinction using one fixed multistream graph backbone and synthetic missing-joint protocols. A 2×2 comparison of global and part supervised contrastive terms is supplemented by part-triplet and ungated-loss controls, giving six objectives trained with three seeds. Input representation, augmentation, batch construction, base identity losses, validation selection, and evaluation masks are shared. A second study freezes the baseline checkpoints and varies the local descriptor and scoring rule. Finally, five runs with a fixed full-training recipe evaluate the selected descriptor on the official CASIA-B test identities. Earlier project experiments examined those identities; this final evaluation assesses within-dataset transfer and is not an independently reserved test.

The study asks whether auxiliary global or part objectives improve retrieval, whether useful local matching requires a trained projection, and whether confidence weighting explains the matching gain. The tested part-SupCon term reduces global-descriptor performance under synthetic corruption. Global SupCon remains inconclusive, and the part-triplet control develops near-collapsed local embeddings. In contrast, matching normalized raw part pools improves clean and corrupted retrieval from the same baseline checkpoints. Uniform and reliability-weighted matching perform similarly, showing that the material gain is associated with local score fusion.

The contributions are:

1. A controlled six-objective comparison separating auxiliary local supervision from retrieval scoring on a shared pose-gait backbone.
2. Frozen-checkpoint controls that remove local projection heads, vary ten projection initializations and five fusion weights, and isolate the incremental effect of reliability weighting.
3. Paired outcome analysis across model seeds and identities for clean, corrupted-probe, and corrupted-gallery endpoints, together with a five-seed standard-protocol evaluation of the selected matcher.

These findings define a useful design check: evaluate a deterministic local matcher before attributing a part-aware system's gain to auxiliary local training. The present evidence establishes that check within this pipeline; replication on established backbones and independent data is needed to assess its generality.

2 Related Work

2.1 Contrastive learning and local identity objectives

SimCLR, MoCo, and supervised contrastive learning emphasize positive-pair construction and projection geometry (T. Chen, Kornblith, Norouzi, & Hinton, 2020; He, Fan, Wu, Xie, & Girshick, 2020; Khosla et al., 2020). Skeleton-specific transformations and representations have supported action recognition and person re-identification (Rao & Miao, 2024; Thoker, Doughty, & Snoek, 2021). ActCLR uses motion-dependent regions for contrastive action learning (L. Lin, Zhang, & Liu, 2023), while GaitPT studies large-scale skeleton gait pretraining (Cosma, Catruna, & Radoi, 2026). These results motivate structured contrastive learning; they do not determine whether an independent identity objective is useful for each anatomical gait region.

Alignment and uniformity provide a language for embedding geometry (T. Wang & Isola, 2020). BYOL and SimSiam examine alternatives to explicit negative pairs (X. Chen & He, 2021; Grill et al., 2020); Barlow Twins and VICReg address redundancy and variance (Bardes, Ponce, & LeCun, 2022; Zbontar, Jing, Misra, LeCun, & Deny, 2021). We use variance diagnostics to identify a failure mode of the tested local triplet objective, without inferring that these alternative objectives would resolve it.

2.2 Pose-gait representations and missing evidence

PoseGait uses body pose and human priors (Liao et al., 2020); GaitGraph and GaitGraph2 learn graph-based gait features (Teepe et al., 2022, 2021). GPGait combines human-oriented normalization with part-aware local–global graph extraction (Fu, Meng, Hou, Hu, & Huang, 2023), and SkeletonGait uses skeleton maps (Fan, Ma, Jin, Shen, & Yu, 2024). OpenGait and FastPoseGait support standardized and reproducible comparisons (Fan et al., 2023; Meng, Fu, Hou, Cao, Liu, and Huang, 2023). Our encoder is an internal control, not a reproduction of those systems.

Related action-recognition encoders use adaptive graphs, action-dependent relations, multi-scale neighborhoods, shift operations, and heatmap volumes (Cheng et al., 2020; Duan, Zhao, Chen, Lin, & Dai, 2022; M. Li et al., 2019; Liu, Zhang, Chen, Wang, & Ouyang, 2020; Shi, Zhang, Cheng, & Lu, 2019). Holding an encoder fixed isolates the intervention within that encoder; replication across these representation choices is a separate question.

Real occlusion combines missing information with incorrect localization and misalignment (P. Huang et al., 2024; T. Li et al., 2024). Occluded skeleton-action work additionally studies imputation (Y. Chen et al., 2023). GREW and Gait3D provide in-the-wild conditions (Zheng et al., 2022; Zhu et al., 2021), while LidarGait examines a different geometric modality (Shen et al., 2023). Our synthetic masks isolate missingness effects and do not reproduce these broader sources of variation.

2.3 Part-aware representations and matching

GaitSet pools silhouette sets, and disentangling methods separate identity from appearance or covariates (Chao, He, Zhang, & Feng, 2019; X. Li, Makihara, Xu, Yagi, & Ren, 2020; Zhang et al., 2019). GaitPart, GaitGL, and 3D Local CNNs extract complementary local and global gait features (Fan et al., 2020; Z. Huang et al., 2021; B. Lin, Zhang, & Yu, 2021). Strip-based, context-sensitive temporal, dynamic-region, and template representations offer further spatial and temporal aggregation choices (P. Huang, Hou, Huang, & Huang, 2025; X. Huang et al., 2021; M. Wang et al., 2023, 2022). GPGait is particularly relevant because its graph structure includes local keypoint relations (Fu et al., 2023). Our narrower comparison holds the representation architecture fixed and asks whether a local metric objective is needed for useful local matching. It does not claim novelty for anatomical pooling or global–local fusion themselves.

2.4 Controlled metric-learning evaluation

Negative sampling and pair weighting influence metric objectives (X. Wang, Han, Huang, Dong, & Scott, 2019; Wu, Manmatha, Smola, & Krahenbuhl, 2017). A Metric Learning Reality Check shows why training and evaluation choices must be aligned when comparing losses (Musgrave, Belongie, & Lim, 2020). Multi-objective optimization and gradient surgery motivate inspecting interactions between losses (Sener & Koltun, 2018; T. Yu et al., 2020). We compare objectives at fixed settings, examine descriptor controls separately, and report gradient agreement as a diagnostic rather than a causal explanation.

3 Controlled Study

Figure 1 summarizes the controlled design. Training changes only the auxiliary objective attached to shared global or anatomical features; the checkpoint-only control study freezes the baseline encoder and changes only the local descriptor and matching rule.

3.1 Data and identity-disjoint evaluation

We use the public HRNet-derived (Sun, Xiao, Liu, & Wang, 2019) 2D pose release for CASIA-B (S. Yu, Tan, & Tan, 2006). Each frame contains 17 joints following the COCO keypoint convention (T.-Y. Lin et al., 2014), with two coordinates and a continuous confidence value. For development, a fixed split seed (2026) holds out identities {002, 006, 012, 025, 026, 032, 043, 046, 054, 059, 061, 072} for validation; the remaining 62 identities train the models. For every held-out identity, normal sequences 01–04 form the gallery and normal sequences 05–06, bag sequences 01–02, and

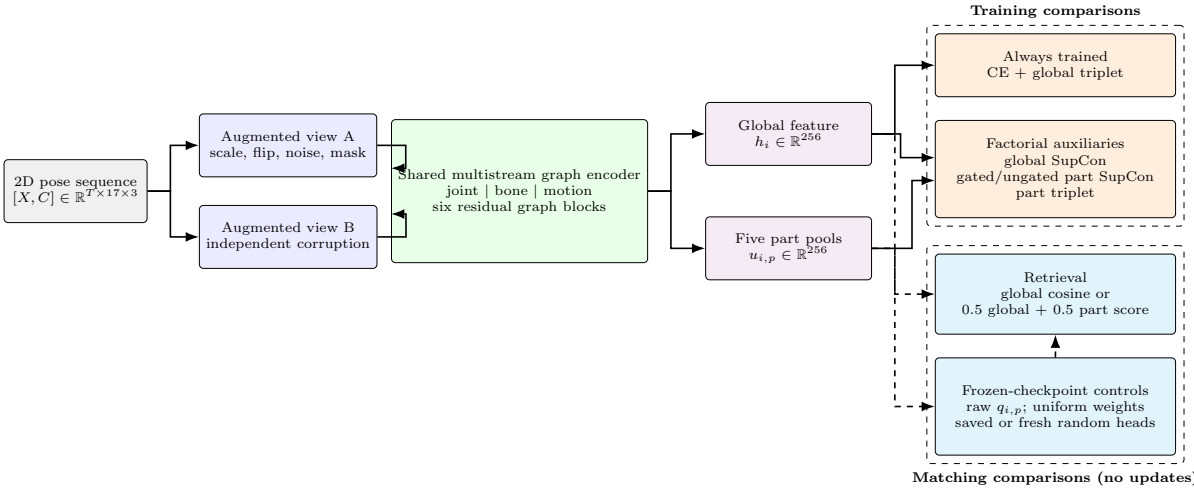


Fig. 1 Separation of part-aware training from part-aware matching. Solid arrows denote the shared forward path. The upper branch contains the six training objectives; the lower branch contains frozen-checkpoint descriptor controls and performs no parameter updates.

clothing sequences 01–02 form the probe set. This gives 792 probe sequences. Identical-view gallery entries are excluded when a probe is scored.

The full-training evaluation uses identities disjoint from the development validation identities. It trains the selected metric baseline on identities 001–074 and evaluates identities 075–124. The standard single-view protocol uses normal sequences 01–04 as gallery and normal 05–06, bag 01–02, and clothing 01–02 as probes, averages over the 11 gallery views, and excludes identical-view matches. The resulting test set contains 3,300 probe sequences from 50 identities. We additionally use a pooled multi-view gallery for the controlled synthetic-corruption endpoints; this pooled protocol is reported separately from the standard single-view score. Earlier exploratory runs in this project evaluated the same identities 075–124. The current development stage excludes them from checkpoint selection, but the project history prevents describing this test population as untouched. An external dataset is required for independent confirmation.

Coordinates are centered on the available pelvis joints, falling back to the visible-joint centroid if the pelvis is missing, and are normalized by sequence body scale. Confidence is supplied as a separate input and pooling weight; it does not multiply and geometrically shrink visible coordinates. Training uses contiguous 60-frame clips, with cyclic repetition for shorter sequences. Evaluation uses every real frame in each sequence.

3.2 Multistream graph backbone

Let $X \in \mathbb{R}^{T\times 17\times 2}$ be a normalized pose and $C \in [0,1]^{T\times 17}$ its confidence. The encoder constructs three input streams: joints $[X, C]$; bones, defined relative to a rooted skeleton parent and observed only when both endpoints are available; and motion at lags one and two, again retaining confidence only when both temporal endpoints are observed. Stream-specific stems feed six residual graph blocks. Each block uses learned perturbations of the physical adjacency and temporal convolutions with dilations 1, 2, and 4. The hidden width grows from 64 to 256, and confidence- weighted global pooling produces a 256-dimensional normalized embedding h_i.

For local representations, we define five overlapping COCO regions: the torso (both shoulders and hips), each arm (shoulder, elbow, wrist), and each leg (hip, knee, ankle). Shoulder and hip joints belong to both the torso and their respective limb; the five face joints are excluded from local pools. Confidence-weighted pooling within part p produces $u_{i,p} \in \mathbb{R}^{256}$. The deterministic raw descriptor is $q_{i,p} = u_{i,p}/\|u_{i,p}\|_2$. The original learned alternative passes $u_{i,p}$ through a part-specific two-layer projection head and ℓ_2

normalization to obtain $z_{i,p}$. Part reliability is

$$r_{i,p} = \frac{1}{T_i|p|} \sum_{t=1}^{T_i} \sum_{j \in p} C_{i,t,j}, \tag{1}$$

where T_i counts real rather than padded frames.

3.3 Shared training recipe

Each batch samples 16 identities and four sequences per identity. Two views of each sequence receive scale, horizontal-flip, and coordinate-noise augmentation. Each view independently has 0.3 probability of remaining geometrically complete; otherwise one of random-joint, body-part, or temporal masking is sampled. The maximum training severity increases linearly from 0.05 to 0.5 over the first half of training. Dynamic-joint corruption is held out from the training sampler and used only for evaluation.

All configurations share the base objective

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{CE,aug}} + 0.5\mathcal{L}_{\text{CE,clean}} + \mathcal{L}_{\text{Tri}}, \tag{2}$$

where the first term classifies both augmented views, the clean anchor classifies the original sequence, and $\mathcal{L}_{\text{Tri}}$ is batch-hard triplet loss (Hermans, Beyer, & Leibe, 2017) on h_i. Same-source augmented views are excluded as triplet positives, so positives must be distinct sequences of the same identity.

Global supervised contrastive loss $\mathcal{L}_{\text{gSC}}$ operates on a projection of the global embeddings and uses every same-identity view as a positive. Part supervised contrastive loss applies the same construction independently to each $z_{i,p}$. In the gated version, a positive pair receives weight $r_{i,p}r_{j,p}$ and the anchor receives weight $r_{i,p}$; parts below reliability 0.05 are excluded from both positives and negatives. The ungated control gives all parts above the same 0.05 reliability threshold unit weight. Part triplet loss $\mathcal{L}_{\text{pTri}}$ applies batch-hard triplet independently to every present part. The complete factorial objective is

$$\mathcal{L} = \mathcal{L}_{\text{base}} + \gamma_g\mathcal{L}_{\text{gSC}} + \gamma_p\mathcal{L}_{\text{pSC}} + \gamma_{pt}\mathcal{L}_{\text{pTri}}. \tag{3}$$

Development training lasts 160 epochs with AdamW, initial learning rate 10^{-3}, weight decay 10^{-4}, five warm-up epochs, cosine decay to 1% of the initial rate, label smoothing 0.1, global SupCon temperature 0.07, part temperature 0.1, triplet margin 0.2, gradient clipping at 5, and automatic mixed precision. A checkpoint is selected every five epochs using only the held-out validation split. The score assigns 75% weight to clean recognition and 25% to the mean of body-part and random-joint recognition at severity 0.3. Selection uses only the global descriptor for every method.

For the official-test evaluation, only the metric baseline is retrained. All 74 official training identities are used for a fixed 85 epochs with the same base objective and optimization settings. Validation is disabled, no official test outcome is used for checkpoint selection, and epoch 85 is evaluated for each of five seeds (11, 22, 33, 44, and 55). The global, uniform raw-part, and reliability-weighted raw-part descriptors are computed from each same-seed checkpoint at the fixed 0.5 local-score weight. The 85-epoch budget is the median selected baseline epoch in the development comparison and was fixed before these full-training runs.

3.4 Retrieval descriptors

The global descriptor uses cosine similarity, $s_g(i,j) = h_i^\top h_j$. The reliability-weighted part descriptor keeps the same global term and averages local cosine scores over jointly reliable parts. Let $v_{i,p}$ denote the local descriptor under evaluation: either the learned projection $z_{i,p}$ or the normalized raw pool $q_{i,p}$. Then

$$s_p(i,j) = \frac{\sum_p r_{i,p}r_{j,p}v_{i,p}^\top v_{j,p}}{\max(\sum_p r_{i,p}r_{j,p}, 10^{-8})}, \tag{4}$$

$$s_{\text{rel}}(i,j) = 0.5s_g(i,j) + 0.5s_p(i,j). \tag{5}$$

Uniform matching replaces each $r_{i,p}$ in the local score by the presence indicator $\mathbf{1}[r_{i,p} > 0]$. If no local part is jointly available, matching falls back to s_g. Global and projected-part descriptors are evaluated for every development checkpoint; raw-part controls are evaluated for the baseline checkpoints. In particular, the metric baseline and global-SupCon configurations give the part heads no direct loss; their part projection weights therefore remain at their seeded initialization, although the shared backbone changes during training.

Table 1 Controlled training objectives. All unlisted terms and all backbone, augmentation, and optimization settings are identical.

Configuration	γ_g	γ_p	γ_{pt}	Part-SupCon gate
Metric baseline	0	0	0	–
Global SupCon	0.2	0	0	–
Part SupCon	0	0.2	0	reliability
Part triplet	0	0	0.2	–
Global + part SupCon	0.2	0.2	0	reliability
Global + part SupCon (ungated)	0.2	0.2	0	presence only

We resolve this confound in a checkpoint-only control study. The three selected metric-baseline backbones are frozen and each sequence is encoded once per corruption realization. We compare (i) $q_{i,p}$ with reliability-weighted matching; (ii) $q_{i,p}$ with uniform weights over jointly present parts; (iii) the saved untrained heads; and (iv) ten freshly initialized frozen copies of the same nonlinear head architecture per backbone checkpoint. No condition updates model parameters. The local-score weight is fixed at 0.5 for the primary comparison. We additionally report weights $\{0, 0.25, 0.5, 0.75, 1\}$ as sensitivity analysis; they are not used to replace the pre-specified primary weight after observing validation performance.

3.5 Occlusion protocols and outcomes

We evaluate random joint-frame removal, temporally smoothed dynamic-joint removal, whole-part removal, and a contiguous temporal interruption at nominal severities $\rho \in \{0, 0.1, 0.3, 0.5, 0.7\}$. Whole-region masks can overlap and overshoot the requested missing-joint count, while temporal smoothing changes the marginal dynamic-joint missingness rate. Thus equal nominal severities do not imply equal removed-joint fractions across families. Masks are applied to normalized poses and isolate missing evidence rather than errors in pose estimation. Five deterministic corruption realizations are averaged at every positive severity. In the clean-gallery protocol only probes are corrupted. In the occluded-gallery protocol gallery and probe sequences receive independently seeded corruption of the same family and severity. We additionally report the CASIA-B-style single-view-gallery average over NM, BG, and CL. Development tables use held-out training identities; the dedicated official-test table uses identities 075–124 and is labeled explicitly.

Figure 2 illustrates the four corruption families, and Figure 3 records where each family appears in training and in the two gallery protocols.

The primary measure is Rank-1 retrieval. Rank-5, Rank-10, mean average precision, equal-error rate, and true-acceptance rates are retained in the saved run artifacts but are secondary to the present diagnosis. Development tables report the mean and sample standard deviation over three training seeds. The main corrupted aggregate weights the four families and four positive severities equally; it summarizes this experimental grid, not a deployment distribution. For paired development comparisons, each saved outcome is a per-probe correctness value, averaged over corruption realizations where applicable. We first average the selected corruption cells within each probe, compute candidate-minus-baseline differences within each of 12 identities and three matched seeds, and then resample seeds and identities as crossed factors with replacement for 100,000 draws. Each draw selects three seeds and 12 identities independently; the same identity sample is shared across the sampled seeds because identities recur in every model run. Candidate–baseline pairing is preserved. Intervals are percentile 95% intervals. They are used to express uncertainty, not to support a confirmatory significance claim across the many exploratory endpoints.

Official-test tables report mean and sample standard deviation over five matched training seeds. For the standard single-view aggregate, only seed-level paired outcomes are retained, so candidate-minus-global uncertainty is a paired t interval over the five seed differences. For pooled multi-view and synthetic-corruption endpoints,

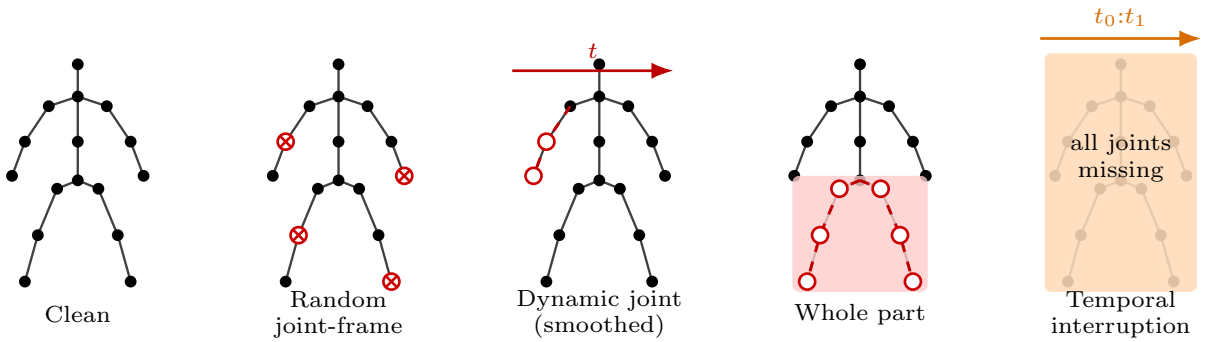


Fig. 2 Conceptual corruption families. Red marks unavailable joints or body regions; orange marks a contiguous interval in which the complete pose is unavailable. Random joint-frame masks vary independently, whereas dynamic-joint masks are temporally smoothed.

Corruption family	Training	Clean gallery	Occluded gallery	Positive ρ
Random joint-frame	sampled	probe only	gallery + probe, independent	5 masks
Dynamic joint	held out	probe only	gallery + probe, independent	5 masks
Whole part	sampled	probe only	gallery + probe, independent	5 masks
Temporal interruption	sampled	probe only	gallery + probe, independent	5 masks

Severities: $\rho \in \{0, 0.1, 0.3, 0.5, 0.7\}$

Descriptors: global and part-aware; primary outcome: Rank-1

Fig. 3 Evaluation matrix. Dynamic-joint corruption is deliberately held out from training. Positive-severity results average five deterministic corruption realizations, and gallery and probe corruptions are independently seeded in the occluded-gallery protocol.

saved per-probe outcomes support a crossed bootstrap: five seeds and 50 test identities are independently resampled with replacement for 100,000 draws while preserving candidate–baseline pairing. Synthetic endpoints first average the four positive severities and four corruption families within each probe.

4 Results

4.1 Part SupCon reduces robustness under the fixed recipe

Table 2 shows the six-objective comparison. With global matching, the metric baseline attains 81.52% clean Rank-1, 63.90% averaged positive-severity clean-gallery Rank-1, and 67.93% when both gallery and probe are corrupted. Global SupCon produces 79.63%, 63.23%, and 67.42%, respectively. The paired intervals in Table 3

Table 2 CASIA-B validation Rank-1 (%). Clean is mean ± seed standard deviation. "All > 0" averages four corruption families at severities 0.1, 0.3, 0.5, and 0.7. Occluded gallery averages the same positive-severity grid. Single-view averages NM, BG, and CL. Every cell uses three training seeds.

Global descriptor					
Training objective	Clean	Clean gallery all > 0	Clean gallery $\rho = 0.5$	Occluded gallery all > 0	Single-view
Metric baseline	81.52 ± 2.43	63.90	64.25	67.93	67.36
Global SupCon	79.63 ± 3.23	63.23	64.15	67.42	66.18
Part SupCon	78.79 ± 1.14	57.78	55.57	64.26	65.33
Part triplet	81.27 ± 3.41	60.65	59.10	66.71	67.82
Global + part SupCon	80.05 ± 1.34	60.70	59.06	65.56	65.75
Global + part SupCon (ungated)	80.89 ± 1.59	61.72	60.59	66.99	66.29

Projected-parts descriptor					
Training objective	Clean	Clean gallery all > 0	Clean gallery $\rho = 0.5$	Occluded gallery all > 0	Single-view
Metric baseline	83.04 ± 1.31	66.31	67.97	70.57	69.34
Global SupCon	82.79 ± 3.36	66.21	67.85	70.38	68.66
Part SupCon	75.42 ± 2.25	54.66	52.65	61.62	61.81
Part triplet	81.31 ± 3.42	60.66	59.11	66.73	67.82
Global + part SupCon	77.86 ± 4.45	57.48	54.84	63.18	63.40
Global + part SupCon (ungated)	78.20 ± 0.07	59.35	58.11	64.85	65.60

include zero for all four global-descriptor endpoints, so this experiment neither demonstrates a benefit nor a clear robustness penalty from the global SupCon term.

Part SupCon behaves differently. Its global descriptor loses 6.12 percentage points on the positive-severity clean-gallery aggregate, with interval $[-10.29, -2.00]$, and 8.68 points at severity 0.5, with interval $[-15.55, -2.38]$. The projected-parts descriptor loses still more. Thus the failure is not limited to a poorly chosen local matching weight: the shared global representation also deteriorates when the local contrastive objective is added.

4.2 Reliability gating is not the primary cause

The combined gated objective reaches 60.70% on the positive-severity global aggregate. Removing reliability weights increases this to 61.72%, and raises the corresponding projected-parts result from 57.48% to 59.35%. The recovery is directionally consistent but remains below the 63.90% and 66.31% metric baselines. Average part reliability (0.502) and presence rate (0.860) are essentially identical across methods because they are determined by the shared input sampler. Gating therefore changes how the same observed evidence weights the auxiliary loss; it does not explain the entire deficit.

4.3 Part-triplet heads exhibit near-collapse

The part-triplet control closely preserves clean global accuracy (81.27%) but falls to 60.65% on the positive-severity aggregate and 59.10% at severity 0.5. Its projected-parts and global results are nearly identical. The training diagnostics in Table 4 are consistent with this behavior: final normalized part variance is approximately 1.7×10^{-6}, and a nonzero finite part–triplet cosine is available in only 9 of 96 scheduled diagnostics. The other diagnostics have a zero-norm gradient for at least one compared loss; they are not numerical failures. Training-log variance at the selected epochs (60, 60, and 40) is also small, averaging 2.03×10^{-6} across seeds and parts. These observations support near-collapse under this particular local objective and training recipe; they do not establish failure of part-triplet learning in general.

Part SupCon avoids variance collapse: its final variance is 3.34×10^{-3}. Nevertheless, its gradient is less aligned with the global triplet objective than the global SupCon gradient is. In the combined

Table 3 Paired Rank-1 difference from the metric baseline in percentage points (crossed 95% bootstrap interval over seed and validation identity). Negative values favor the baseline.

Training objective	Descriptor	Clean	Clean gallery, all > 0
Global SupCon	Global	-1.89 [-5.30, +1.05]	-0.67 [-4.86, +3.79]
Part SupCon	Global	-2.74 [-6.69, +1.18]	-6.12 [-10.29, -2.00]
Part triplet	Global	-0.25 [-5.22, +4.17]	-3.26 [-7.59, +1.06]
Global + part SupCon	Global	-1.47 [-5.85, +3.03]	-3.21 [-8.97, +2.89]
Global + part (ungated)	Global	-0.63 [-5.35, +5.30]	-2.19 [-5.35, +1.11]
Global SupCon	Projected parts	-0.25 [-3.79, +2.99]	-0.10 [-3.27, +3.08]
Part SupCon	Projected parts	-7.62 [-11.91, -3.03]	-11.65 [-16.02, -7.32]
Part triplet	Projected parts	-1.73 [-6.82, +2.90]	-5.65 [-9.85, -1.16]
Global + part SupCon	Projected parts	-5.18 [-10.61, +0.13]	-8.83 [-14.35, -2.29]
Global + part (ungated)	Projected parts	-4.84 [-8.75, -1.05]	-6.96 [-11.01, -3.08]

Training objective	Descriptor	Clean gallery, $\rho = 0.5$	Occluded gallery, all > 0
Global SupCon	Global	-0.10 [-6.13, +6.76]	-0.51 [-2.90, +1.76]
Part SupCon	Global	-8.68 [-15.55, -2.38]	-3.67 [-6.00, -1.27]
Part triplet	Global	-5.15 [-11.42, +1.28]	-1.22 [-4.29, +1.50]
Global + part SupCon	Global	-5.19 [-13.69, +4.17]	-2.37 [-6.16, +1.24]
Global + part (ungated)	Global	-3.65 [-8.91, +0.68]	-0.94 [-3.00, +1.05]
Global SupCon	Projected parts	-0.12 [-4.37, +4.10]	-0.20 [-2.13, +1.76]
Part SupCon	Projected parts	-15.32 [-22.19, -8.96]	-8.96 [-11.51, -6.03]
Part triplet	Projected parts	-8.86 [-15.80, -1.57]	-3.85 [-6.48, -1.32]
Global + part SupCon	Projected parts	-13.13 [-20.45, -4.51]	-7.39 [-11.19, -2.97]
Global + part (ungated)	Projected parts	-9.85 [-16.07, -4.53]	-5.73 [-8.08, -3.59]

model, the mean part–triplet cosine is 0.203, compared with a global–triplet cosine of 0.446. These cosines are positive, so the data do not support a claim of direct gradient opposition. They instead show weaker agreement and leave open other causes, including excessive local invariance, part-specific identity ambiguity, or mismatch between the local contrastive task and retrieval.

4.4 Raw part matching resolves the random-head confound

Table 5 removes the learned projection from the same checkpoints. Reliability-weighted raw parts improve clean Rank-1 by 2.95 points over global matching, with crossed paired interval $[1.05, 5.30]$, and improve the positive-severity aggregate by 4.32 points, $[2.89, 5.71]$. The gain grows to 6.06 points at severity 0.5, $[3.65, 8.59]$, and remains 3.83 points with independently corrupted galleries, $[2.79, 5.10]$. Every training seed improves on both clean and corrupted-probe endpoints, and the four family-specific corrupted aggregates all increase.

All ten fresh random-head seeds also pass the clean and robustness thresholds: their clean gains range from 1.68 to 2.74 points and their corrupted-probe gains from 2.42 to 2.82 points. Thus the original improvement did not depend on one favorable initialization. However, removing the nonlinear projection is both deterministic and stronger. The effect is also insensitive to moderate fusion: raw reliable parts improve the corrupted aggregate by 2.30, 4.32, and 5.68 points at weights 0.25, 0.5, and 0.75, respectively. We retain 0.5 as the declared primary weight rather than selecting 0.75 after observing validation.

Reliability weighting is not the main source of the raw-descriptor gain. Relative to uniform averaging, it changes clean Rank-1 by only $+0.04$ points and the principal corrupted aggregate by $+0.12$ points. The crossed interval for the latter difference, $[-0.02, +0.24]$, includes zero. The small mean difference arises mainly under whole-part corruption. These comparisons associate the material gain with regional pooling and score

Table 4 Training diagnostics averaged over three seeds. Part variance is the mean coordinate-wise variance over the final 20 epochs. Cosines are measured every five epochs; parentheses give nonzero finite cosine measurements out of 96 scheduled.

Objective	Variance ($\times 10^{-3}$)	Part–global	Part–triplet
Metric baseline	1.269	–	–
Global SupCon	1.300	–	–
Part SupCon	3.338	–	0.212 (95)
Part triplet	0.002	–	0.012 (9)
Global + part SupCon	3.349	0.265 (96)	0.203 (96)
Global + part SupCon (ungated)	3.242	0.241 (96)	0.195 (96)

Table 5 Checkpoint-only descriptor controls on CASIA-B validation Rank-1 (%). All conditions use the same three selected metric-baseline backbones and the fixed 0.5 fusion weight. "Fresh random heads" is the mean across ten projection seeds; brackets give the range after averaging the three backbone seeds. No control retrains the backbone or local head.

Descriptor	Clean	Clean gallery all > 0	Clean gallery $\rho = 0.5$	Occluded gallery all > 0
Global only	81.52	63.90	64.25	67.93
Saved frozen head	83.04	66.31	67.97	70.57
Fresh random heads	83.75 [83.21, 84.26]	66.50 [66.32, 66.73]	68.07 [67.71, 68.40]	70.70 [70.58, 70.81]
Raw parts, uniform	84.43	68.11	69.99	71.71
Raw parts, reliability weighted	**84.47**	**68.23**	**70.31**	**71.76**

fusion; random-region controls would be needed to attribute it specifically to anatomical grouping.

Figure 4 disaggregates the corruption endpoint by family and severity. The curves show where global matching, the auxiliary objectives, and raw-part fusion differ, including the dynamic family omitted from training. An aggregate improvement should therefore be interpreted alongside these family-specific accuracy profiles.

4.5 Official CASIA-B test evaluation

Table 6 evaluates the selected matcher without changing the selected objective, descriptor, or fusion weight. On the standard protocol, reliability-weighted raw parts improve the three-condition average over global matching by 5.69 percentage points; the paired five-seed 95% interval is $[5.19, 6.20]$. The condition-specific gains are 6.50 points for NM, $[6.16, 6.84]$; 5.97 for BG, $[5.73, 6.22]$; and 4.61 for CL, $[3.51, 5.71]$.

The saved test-identity outcomes additionally permit identity resampling for the pooled protocol. Reliability-weighted raw parts gain 6.58 points on uncorrupted probes, with crossed-bootstrap interval $[5.37, 7.83]$; 6.39 points over the positive- severity synthetic grid with a clean gallery, $[5.40, 7.40]$; and 5.63 points with independently corrupted galleries, $[4.89, 6.40]$. Uniform and reliability- weighted raw matching remain practically tied on the standard average: the reliability-minus-uniform difference is $+0.004$ points with paired interval $[-0.059, +0.067]$. Reliability gives small positive synthetic-grid differences of 0.16 points for the clean gallery and 0.08 points for the occluded gallery, while the material gain remains associated with adding local scores. The standard-protocol gain is positive in all five paired runs, ranging from 5.05 to 6.02 points. This consistency establishes the effect within the evaluated system, despite prior project exposure to the test identities.

4.6 Efficiency

The common backbone contains 1.69 million parameters. Mean training time is 1.63 hours for the metric baseline, 1.89 hours for global SupCon, and 1.97–1.99 hours for the combined or part-triplet variants on the recorded H100 MIG environment. This is approximately 21% overhead for the combined model. These measurements do not establish deployment cost: the raw-part descriptor stores and scores five additional local embeddings, giving 1,536 feature coordinates per sequence instead of 256, plus reliability values.

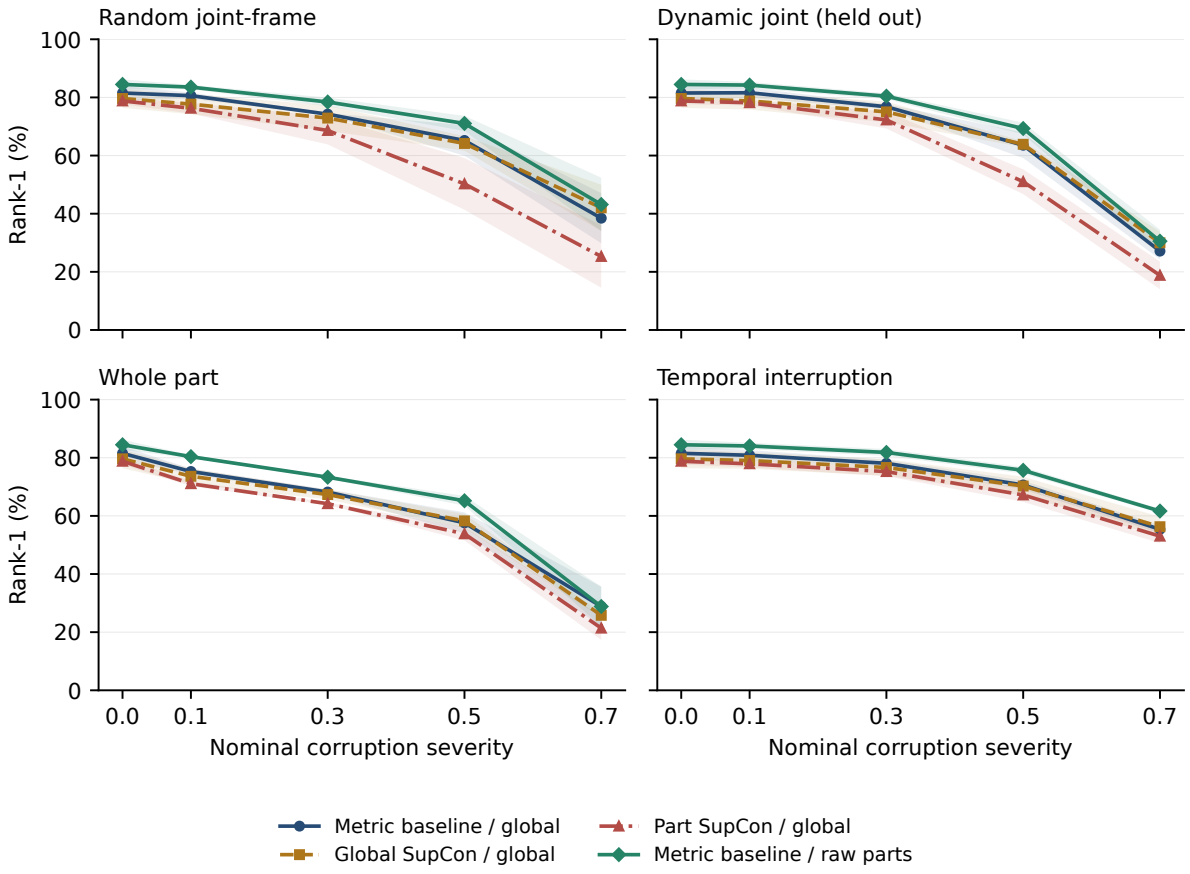


Fig. 4 CASIA-B development severity curves with a clean pooled multi-view gallery. Lines are means over three training seeds; shaded bands show one sample standard deviation across seeds, not confidence intervals. Positive severities average five deterministic mask realizations. The raw-part matcher uses the same metric-baseline checkpoints at fusion weight 0.5. Severity is nominal and is not calibrated to equal missing-joint fractions across families.

This is a sixfold increase in uncompressed feature storage at equal precision. Inference latency and pairwise scoring throughput were not measured; parameter-free matching does not imply cost-free retrieval.

5 Discussion

5.1 Interpreting training and matching separately

The clearest supported distinction is between a local feature's usefulness as complementary retrieval evidence and the effect of an independent identity loss on that feature. With the tested settings, part SupCon reduces corrupted-probe accuracy even when retrieval uses only the global descriptor. Its penalty therefore extends beyond the projected local matcher. Removing gating partly recovers performance, but does not restore the baseline. Global SupCon remains inconclusive, while part triplet exhibits near-collapse. These results concern the tested auxiliary terms, weights, temperatures, and shared selection rule; they do not establish that local supervision is generally harmful.

Frozen-checkpoint controls identify a constructive alternative. Raw regional pools improve retrieval without a separate local objective or learned local projection. Ten fresh projection seeds also help, reducing the likelihood that one favorable initialization explains the saved-head

Table 6 Five-seed CASIA-B official-test Rank-1 (%), reported as mean ± sample standard deviation. The upper panel follows the standard single-view-gallery protocol and excludes identical views. The lower panel uses the separately labeled pooled multi-view gallery; "synthetic" averages four corruption families at severities 0.1, 0.3, 0.5, and 0.7.

Standard single-view protocol Descriptor	NM	BG	CL	Average
Global only	67.64 ± 2.83	49.50 ± 1.92	27.91 ± 0.94	48.35 ± 1.85
Raw parts, uniform	74.02 ± 3.04	55.56 ± 2.04	32.54 ± 1.88	54.04 ± 2.21
Raw parts, reliability weighted	74.14 ± 3.04	55.47 ± 2.01	32.52 ± 1.81	54.04 ± 2.18

Pooled multi-view controlled endpoints Descriptor	Uncorrupted	Synthetic, clean gallery	Synthetic, occluded gallery
Global only	68.06 ± 0.50	48.90 ± 0.63	54.34 ± 0.63
Raw parts, uniform	74.76 ± 0.47	55.13 ± 1.22	59.88 ± 0.90
Raw parts, reliability weighted	74.64 ± 0.45	55.29 ± 1.22	59.96 ± 0.90

result. Uniform matching performs almost identically to reliability weighting. Consequently, confidence gating is not supported as a substantial independent contribution. The experiment establishes an association between regional score fusion and better retrieval, rather than proving that the anatomical grouping itself is optimal. Random groups matched in size and dimension remain a necessary mechanism control.

The resulting design check is operational: compare global matching, raw regional matching, and projected regional matching on the same frozen checkpoint before attributing a system-level gain to its local training loss. This check can expose a useful descriptor already present in the globally trained feature map. Whether it generalizes is an empirical question for other backbones and data.

5.2 Scope of the official-test result

The fixed 0.5 raw-part matcher retains its gain across five full-training runs and all three standard CASIA-B conditions. It does not establish the six-way objective ranking on official test identities: only the selected metric baseline is retrained in that stage. Further, those test identities were examined in earlier exploratory project experiments. The result provides within-dataset evidence for the matching decision, with the recipe frozen before the current runs; independent confirmation requires external data.

Absolute performance also constrains relevance. FastPoseGait reports HRNet-based standard-protocol averages of 71.83% for GaitGraph2 and 76.04% for GaitGraph (Meng et al., 2023), above this pipeline's 54.04% raw-matcher result. These literature values use their own training recipes and are contextual rather than controlled comparisons. Reproducing an established system and repeating the descriptor intervention would test whether this finding survives a more competitive representation. Real occlusion and an independent dataset would then test whether the observed synthetic-missingness benefit extends to detector errors and environmental variation.

6 Limitations and Validity Threats

Prior test exposure and single-dataset evidence.

Earlier project experiments evaluated identities 075–124. Although the current development stage uses only held-out training identities for selection, the official split cannot be regarded as an untouched project-level holdout. The five-seed result is a fixed-recipe within-dataset evaluation. It covers the selected baseline and matchers, while the negative objective comparison remains a development result on 12 identities.

One encoder and fixed objective settings.

All runs use one HRNet-derived pose source and an internal multistream graph encoder. The six-way comparison holds loss weights and temperatures fixed; method-specific tuning, alternative part objectives, and other established backbones may change the ranking. Global-only checkpoint

selection provides a common rule but may favor a different epoch from selection by local matching. These controls isolate behavior under a common recipe, not each objective's best attainable accuracy.

Synthetic missingness.

Masks are applied after pose normalization and do not reproduce left/right swaps, cross-person joint assignment, plausible but incorrect tracks, or occlusion-induced detector and normalization errors. Nominal severity differs from actual removed-joint fraction for whole-part and smoothed dynamic masks. The equally weighted corruption aggregate is a study endpoint rather than an estimate of expected operational accuracy.

Incomplete mechanism controls.

Raw matching is evaluated on baseline checkpoints rather than on all six trained objectives. Random regions and feature-dimension controls are absent, so the independent role of anatomy is unresolved. Gradient cosines are measured at the shared paired-view feature tensor; the clean-CE branch occupies a different slice and is orthogonal there by construction. Positive cosines indicate agreement at that tensor, not parameter-level causality. Variance diagnostics describe training batches rather than test embedding distributions.

Sampling and model development.

Three development seeds and 12 identities limit uncertainty estimation; the crossed bootstrap preserves their shared identity structure but cannot create additional independent training runs. The official standard-protocol paired t interval uses five seeds and four degrees of freedom and represents training-seed variability rather than test-population uncertainty. Many corruption and descriptor comparisons are exploratory; no family-wise error claim is made. Descriptor choice uses the same development identities as the controls and fusion sensitivity. The declared 0.5 weight remains fixed, and higher development scores at 0.75 are not promoted after inspection.

Provenance and retrieval cost.

The factorial records source SHA-256 "ad96b902…" and the descriptor controls "eddf23a2…", both with dataset SHA-256 "123e838c…". Official runs record source "7a351cac…", dataset "9823d021…", and configuration fingerprint "f25bd174…". Configurations, histories, metrics, environment records, and probe outcomes are retained; the development reanalysis includes an artifact manifest and deterministic bootstrap settings. The runs nevertheless record "not-a-git-checkout" rather than immutable repository commits. Archiving the matching source snapshots and locked dependencies remains a release requirement. Five official runs use two H100 NVL MIG profiles with matching software, source, data, and configuration hashes. Retrieval latency and throughput are unmeasured, and the raw matcher stores six feature vectors rather than one.

7 Ethical Considerations

Gait is biometric information and can enable remote identification without an individual's awareness. Improvements in occlusion handling do not remove the requirements for lawful purpose, consent where applicable, access control, retention limits, and human oversight. Evaluation should include demographic and disability-related audits, including mobility aids and atypical gait. The present CASIA-B data cannot support fairness or deployment claims, and the reported synthetic robustness should not be interpreted as operational reliability.

8 Conclusion

This controlled study separates local identity supervision from local matching in one pose-gait pipeline. Under the fixed recipe, part SupCon reduces the principal synthetic-corruption endpoint by 6.12 percentage points, while raw part matching improves it by 4.32 points from the same baseline checkpoints. Crossed paired intervals exclude zero for both effects. Random-head controls and nearly tied uniform and reliability-weighted scores associate the matching gain with regional score fusion rather than a favorable projection seed or a substantial confidence-gating benefit.

Five full-training runs preserve the matching gain on the standard CASIA-B protocol: average Rank-1 increases from $48.35 \pm 1.85\%$ to $54.04 \pm$

2.18%, a paired gain of 5.69 points with interval [5.19, 6.20]. Earlier project exposure to these test identities limits independent confirmation. The practical implication is to evaluate deterministic local matching separately before crediting an auxiliary part loss. Its broader relevance depends on replication with established backbones, independent datasets, and real-occlusion evidence.

Statements and Declarations

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Competing interests

The authors declare that they have no competing interests.

Data availability

This study uses the HRNet-derived CASIA-B pose release distributed through FastPoseGait (https://github.com/BNU-IVC/FastPoseGait), with the dataset available from Science Data Bank at https://www.scidb.cn/en/detail?dataSetId=8ec62efd66a544939e821edeccc1f35c. Access and reuse remain subject to the original providers' terms; dataset files are not redistributed with this work. The identity splits, preprocessing, corruption protocols, and statistical analyses are described in this article.

Code availability

The implementation, experiment configurations, SLURM workflows, and aggregation utilities are available at https://github.com/likwifi/oap-supcon.

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